



3 Volt Intel[®] Fast Boot Block to Hitachi SH7750 (SH-4) CPU Design Guide

Application Note 712

April 2000



Information in this document is provided in connection with Intel® products. No license, express or implied, by estoppel or otherwise, to any intellectual property rights is granted by this document. Except as provided in Intel's Terms and Conditions of Sale for such products, Intel assumes no liability whatsoever, and Intel disclaims any express or implied warranty, relating to sale and/or use of Intel products including liability or warranties relating to fitness for a particular purpose, merchantability, or infringement of any patent, copyright or other intellectual property right. Intel products are not intended for use in medical, life saving, or life sustaining applications.

Intel may make changes to specifications and product descriptions at any time, without notice.

Designers must not rely on the absence or characteristics of any features or instructions marked "reserved" or "undefined." Intel reserves these for future definition and shall have no responsibility whatsoever for conflicts or incompatibilities arising from future changes to them.

Contact your local Intel sales office or your distributor to obtain the latest specifications and before placing your product order.

Copies of documents which have an ordering number and are referenced in this document, or other Intel literature may be obtained by calling 1-800-548-4725 or by visiting Intel's website at <http://www.intel.com>.

Copyright © Intel Corporation, 1999-2000

*Other brands and names are the property of their respective owners.

Contents

1.0	Introduction	1
2.0	Hardware Interface	1
3.0	Interfacing the Fast Boot Block Memory to SH7750 at 66 MHz.....	1
	3.1 Interface Considerations	2
	3.2 Processor Interface Signals	2
	3.3 Control Signal Generation	3
4.0	Summary.....	5
A	Additional Information	6

Revision History

Date of Revision	Version	Description
07/20/99	-001	Original version
03/30/00	-002	Reformatted document

1.0 Introduction

3 Volt Intel® Fast Boot Block memory provides low-power, cost-effective, high performance asynchronous page-mode and synchronous burst read operations. Its enhanced architecture and interface dramatically increase read performance beyond previously attainable levels at 8- and 16-Mbit densities. With zero wait-state burst reads up to 54 MHz, the 3 Volt Fast Boot Block is an ideal memory solution for burst CPUs.

This application note will cover the 3 Volt Fast Boot Block memory's interface to the burst microprocessor. It will also discuss some of the general concepts involved when interfacing the 3 Volt Fast Boot Block memory's integrated features and control signals.

2.0 Hardware Interface

The 3 Volt Fast Boot Block memory integrates several new features that increase its CPU interface compatibility to a range of burst microprocessors and microcontrollers:

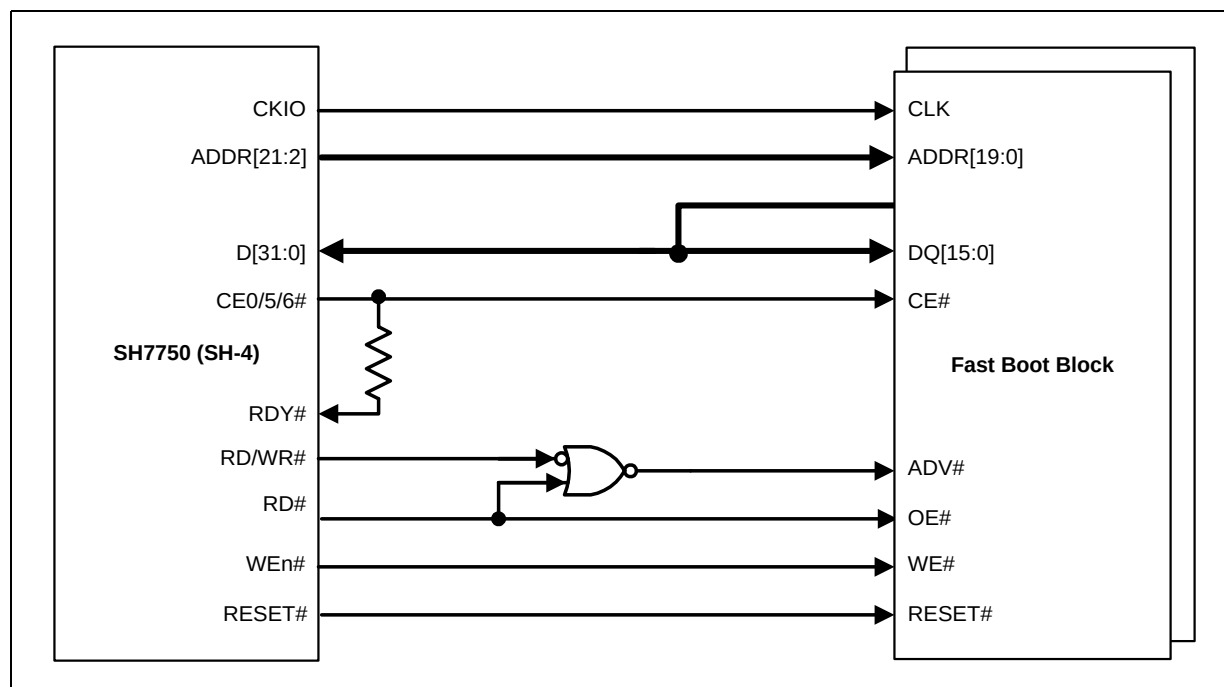
- Address latch
- Read configuration register
- Burst address generator
- WAIT# output signal

The address latch latches the address during read and write cycles. CPUs that employ a multiplexed address/data bus can leverage the internal latch to de-multiplex the bus. ASICs can also use the latch to reduce pin count while improving performance. The read configuration register optimizes the flash memory component to specific CPU characteristics, such as burst length, burst order, operating frequency, and clock configuration. This register tells the burst address generator how to generate addresses during burst-mode reads. Generating the addresses internally eliminates the memory's dependence on the CPU for new addresses during a burst read, which improves system read performance. Finally, a WAIT# output signal is provided to ease CPU-to-flash memory communication and synchronization during continuous burst mode. When configured for continuous bursting, this signal is used as an input to the system wait-state generator. It informs the wait-state generator when a delay is needed during a continuous burst access. Combined, these integrated features enable a simple, and in many cases, "glueless" interface.

3.0 Interfacing the Fast Boot Block Memory to SH7750 at 66 MHz

The SH7750 (SH-4) embedded microprocessor produced by Hitachi employs a 32-bit RISC architecture which features object code upward-compatible with the SH-1, SH-2, and SH-3 microprocessors. The SH-4 uses a 16-bit fixed-length instruction set, which enables program code size to be reduced by about 50%. Figure 1 illustrates the block diagram of the 3 Volt Fast Boot Block memory interface to the SH7750.

Figure 1. The 3 Volt Intel® Fast Boot Block Memory Interface to the SH7750 Microprocessor



3.1 Interface Considerations

The interface uses two Fast Boot Block memory components to match the SH7750's 32-bit data bus width for burstable memory space in area 0, 5, or 6 of the CPU. An OR gate is required in this interface to generate ADV# from the CPU's RD/WR# and RD# signals. A pull-down resistor is required in this interface to control the Ready (RDY#) pin of the CPU. RDY# needs to be pulled low by CE# during flash accesses to indicate the use of the CPU's internal programmable wait-states. The settings for RCR[15:0] are 0010101x10000010.

3.2 Processor Interface Signals

The interface uses the following signals provided by the SH7750.

A₂₁₋₂: The address bus transmits instruction and data addresses to memory and subsystem peripherals.

D₃₁₋₀: The data bus contains bi-directional data paths to transfer data between the processor

CKIO: The Clock signal is used to synchronize all and external memory. processor/memory requests.

CSn#: The Chip Select signal indicates which memory bank the processor is accessing.

RD/WR#: The Read/Write signal indicates whether the current access is a read or a write.

RD#: The Read signal indicates the current access is a read.

WE#: The Write Enable indicates a memory write access.

RDY#: The Ready pin is driven by external devices to insert external wait-states.

3.3 Control Signal Generation

In this interface, the SH7750 directly controls the 3 Volt Fast Boot Block memory CE# with the CSn# signal. OE# and WE# are driven by the RD# and WEn# signals respectively. ADV# is generated by ORing an inverted RD/WR# signal with the RD# signal. ADV# toggles once at the start of read accesses and remains low during writes. Wait-states are generated internally by programming the CPU's respective Wait Control Register. The Wait Control settings for the first cycle should be set for six inserted wait-states, and the Burst Pitch should be set for zero wait-states. A pull-down resistor is connected between CSn# and RDY# to pull RDY# low during flash accesses and to allow other devices to drive RDY# during other memory accesses. The specific value of the resistor will depend on the specific system design.

Figure 2 shows an initial read from following a reset. The rising edge of the processor's RESET# signal indicates a restart to the external peripherals. During this restart, the processor will exit the reset mode 30 clocks after the rising edge of RESET#. Note that the first access after reset is a single read with an initial 15 wait-states inserted by the processor. Because the 3 Volt Fast Boot Block memory is asynchronous after reset, burst reads are allowed only after an initial write operation which synchronizes the flash memory.

Figure 2. The 3 Volt Intel® Fast Boot Block Memory/SH7750 Power-Up Single Read at 66 MHz

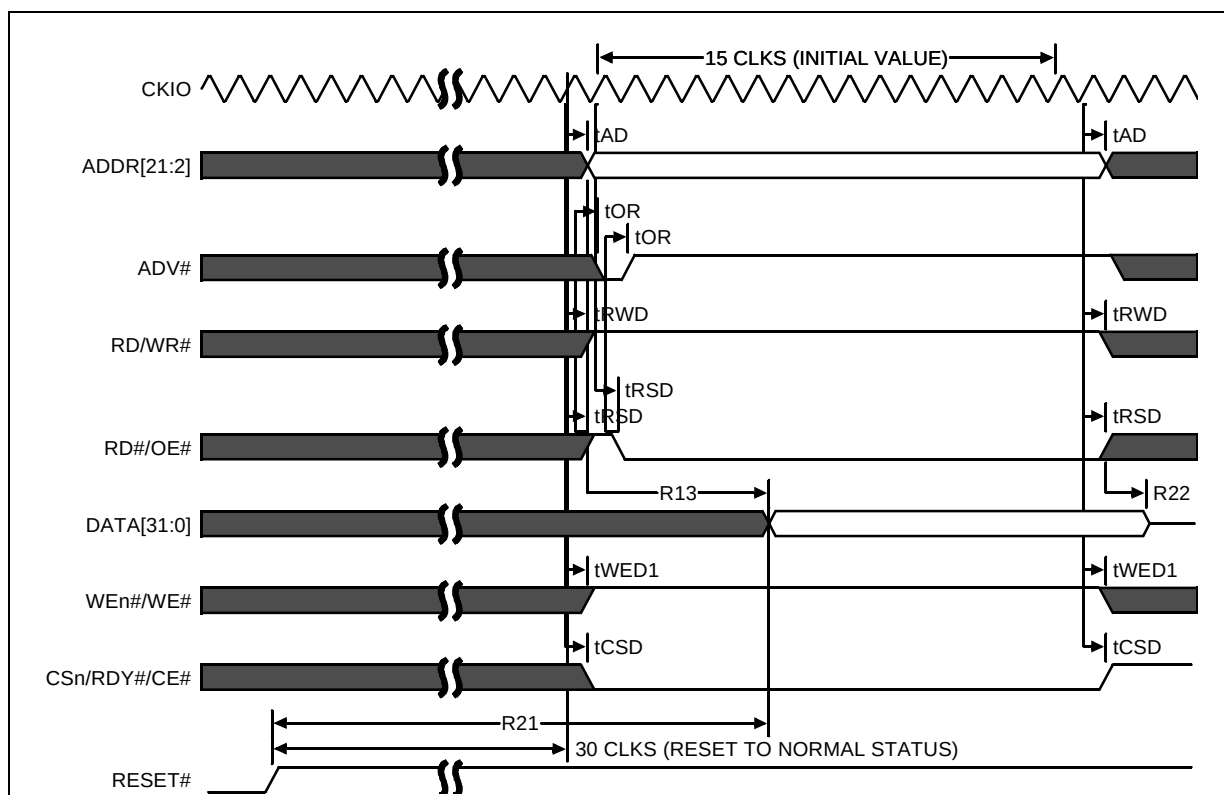


Figure 3 shows a burst read example which contains eight words. Addresses are latched at the falling edge of CKIO by the flash. Note that the processor outputs the address and latches data on the rising clock edge. This configuration was chosen to improve performance. After an initial delay, the 3 Volt Fast Boot Block memory provides zero wait-state performance with a new address at every other clock. Note that the processor's bus cycles are two clocks long. The processor's RD# signal drives OE#. WE# must be held high throughout the read cycle.

Figure 3. The 3 Volt Intel® Fast Boot Block Memory/SH7750 Eight-Word Burst Read Cycle at 66 MHz

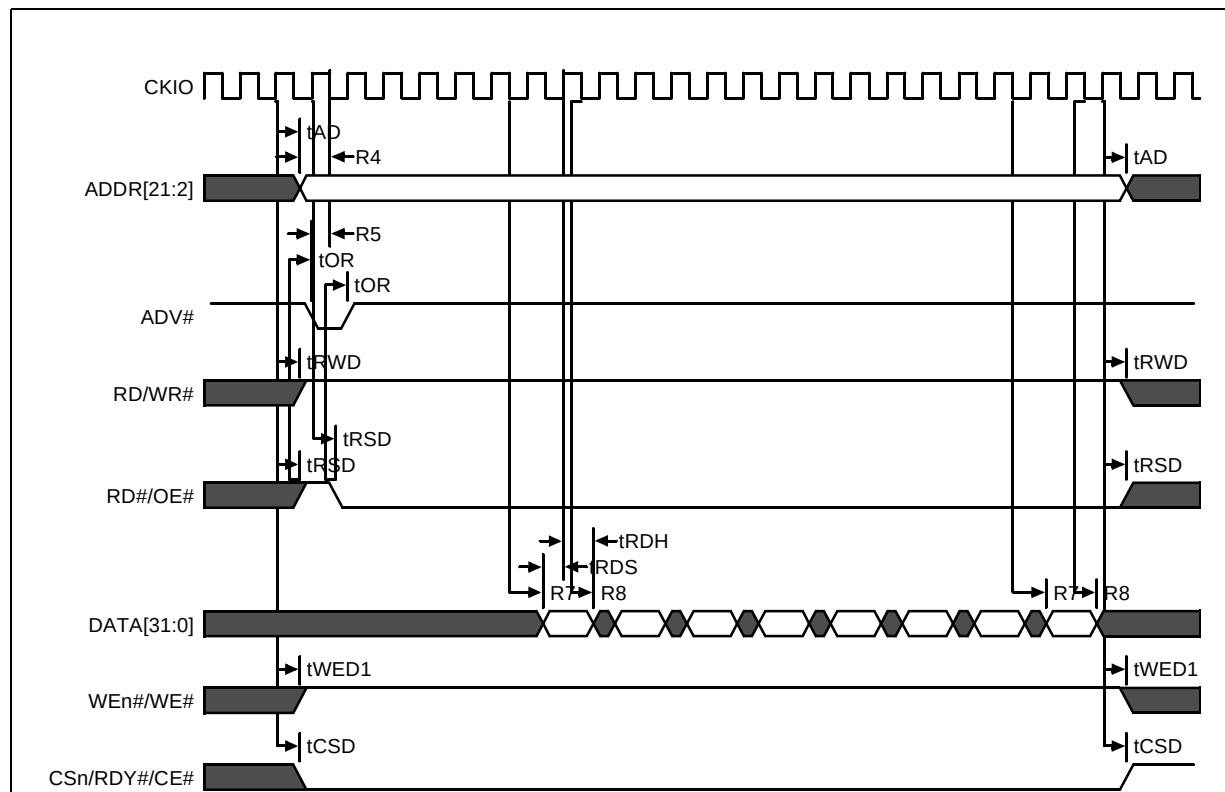
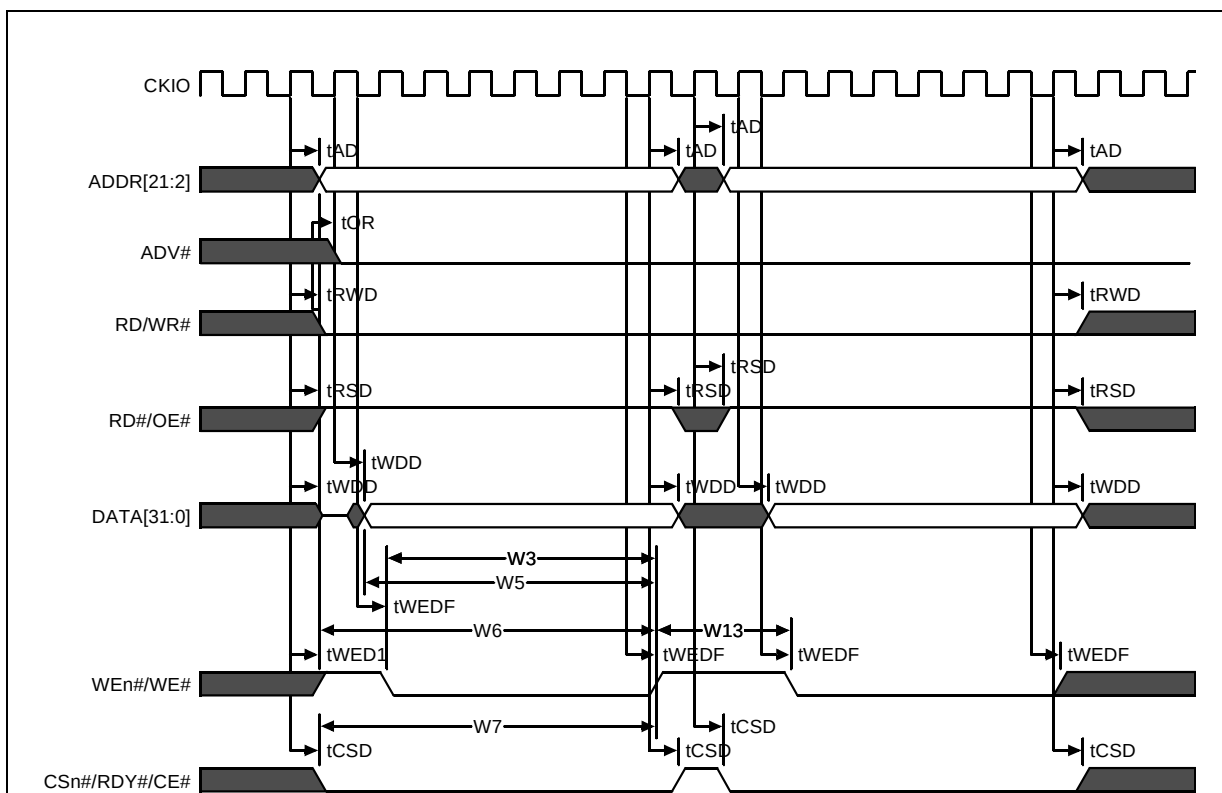


Figure 4 shows a write cycle. The processor's $WEn\#$ signal can be directly used as $WE\#$. $WE0\#$ is used to control the lower 16 data bits on one flash, and $WE2\#$ is used to control the upper 16 data bits on the other flash.

Figure 4. The 3 Volt Intel® Fast Boot Block Memory/SH7750 Write Cycle at 66 MHz



Consult the appropriate datasheets for specific information about the individual components in this interface. Refer to the *SH7750 (SH-4) Datasheet* for timings shown in the waveforms that start with “t.” Refer to the *3 Volt Fast Boot Block Flash Memory; 28F800F3, 28F160F3* datasheet for timings that start with “R” or “W.”

4.0 Summary

The 3 Volt Intel Fast Boot Block memory’s integrated features such as the built-in address latch, the read configuration registers, and the integrated address generator enable simple, and in many cases, “glueless” interfaces to burst processors. These features minimize interface logic while providing high read performance. SmartVoltage capabilities provide fast factory programming and reduce power consumption. A selection of different densities and packages increase design flexibility. With its enhanced architecture and available options, the 3 Volt Fast Boot Block memory provides the best cost/performance value available. The 3 Volt Fast Boot Block memory is the ideal memory solution for burst processors.

Appendix A Additional Information

Order Number	Document/Tool
290644	<i>3 Volt Fast Boot Block Flash Memory; 28F800F3, 28F160F3 datasheet</i>
297939	<i>3 Volt Fast Boot Block Flash Memory; 28F800F3, 28F160F3 Specification Update</i>
292213	<i>AP-655 Fast Boot Block Design Guide</i>
292172	<i>AP-617 Additional Flash Data Protection Using V_{PP}, RP#, and WP#</i>

NOTES:

1. Please call the Intel Literature Center at (800) 548-4725 to request Intel documentation. International customers should contact their local Intel or distribution sales office.
2. Visit Intel's World Wide Web home page at <http://www.Intel.com> for technical documentation and tools.
3. For the most current information on Intel Fast Boot Block products, visit our website at <http://developer.intel.com/design/flash/bblock>